KHULNA
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Course Code : EE 4121

VLSI

Stick Diagram

Stick Diagram:

A pictorial view of the circuit in terms of lines and interconnections required on each MASK level.

- An intermediate state to translate circuit into MASK
- No exact geometric information is known.
- Area can be estimated.
- Lines and Sticks instead of conventional symbols.

Stick Diagrams :

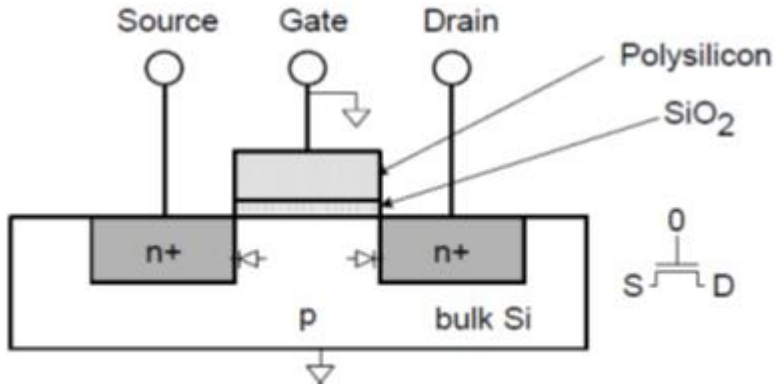
- A stick diagram is a symbolic representation of a layout.
- In stick diagram, each conductive layer is represented by a line of distinct color.
- Width of line is not important, as stick diagrams just give only wiring and routing information.
- Does show all components/vias, relative placement.
- Does not show exact placement, transistor sizes, wire lengths, wire widths, tub boundaries.

Colour Codes :

Layer	Color	Representation
N+ Active	Green	
P+ Active	Yellow	
PolySi	Red	
Metal 1	Blue	
Metal 2	Magenta	
Contact	Black	X
Buried contact	Brown	X
Via	Black	X
Implant	Dotted yellow	
N-Well	Dotted Green/Black	

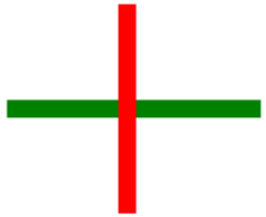
Darshana Sankhe

Enhancement-Mode MOSFET :



Stick Diagrams: Basic Rules

- Poly crosses diffusion forms transistor
- Red (poly) over Green(Active), gives a FET.



nFET/
nMOS

- Aspect Ratio



pFET/pMOS

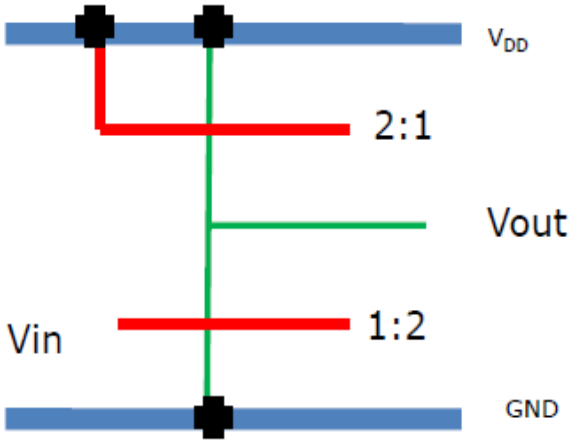
Stick Diagrams: Basic Rules

- Blue may cross over red or green, without connection.
- Connection between layers is specified with **X**.
- Metal lines on different layer can cross one another, connections are done using via.

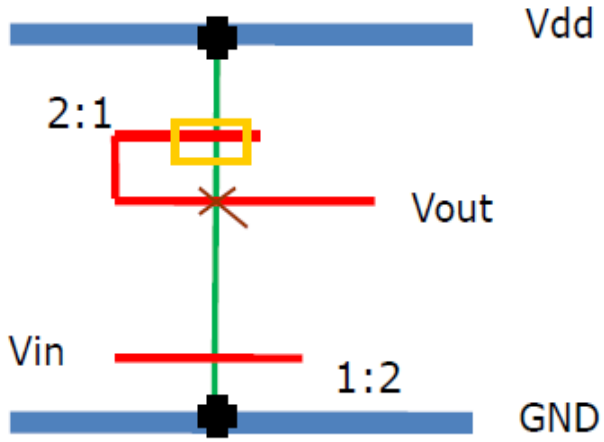
Stick Diagrams(NMOS): Basic Steps

- Normally, the first step is to draw two parallel metal (blue) V_{DD} and GND rails.
- There should be enough space between them for other circuit elements.
- Next, Active (Green) paths must be drawn for required transistors.
- Do not forget to mark contacts as **X**, wherever required.
- Remember, Poly (Red) crosses Active (Green/yellow), where transistor is required.
- For depletion mode FET, draw required implants (yellow).
- Label each transistor with L:W ratio.

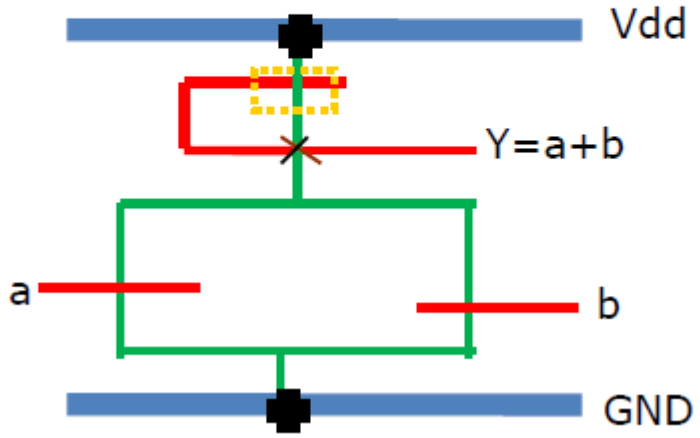
NMOS Inverter: Enhancement load (Stick diagram)



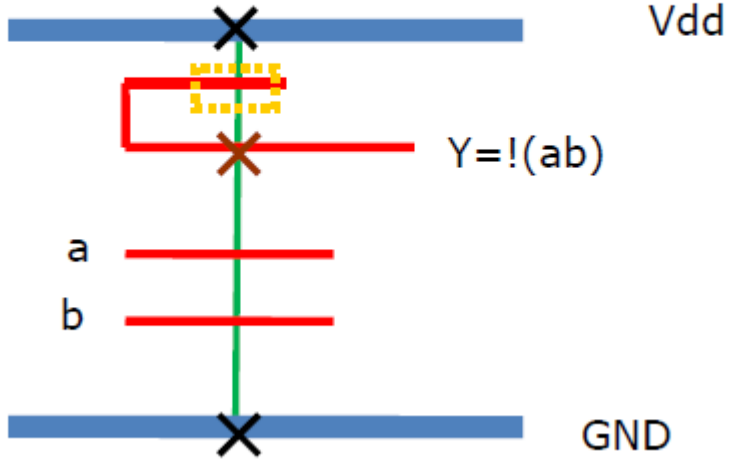
NMOS Inverter: Depletion load (Stick Diagram)



NMOS NOR (Stick Diagram)



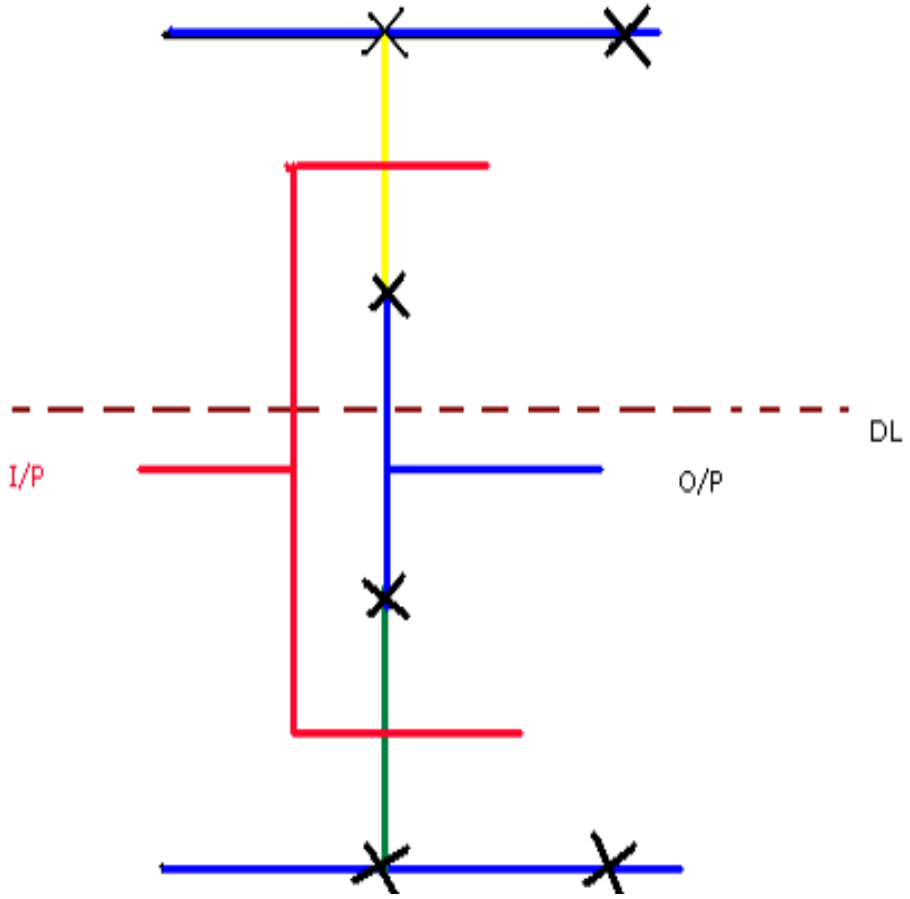
NMOS NAND (Stick Diagram)



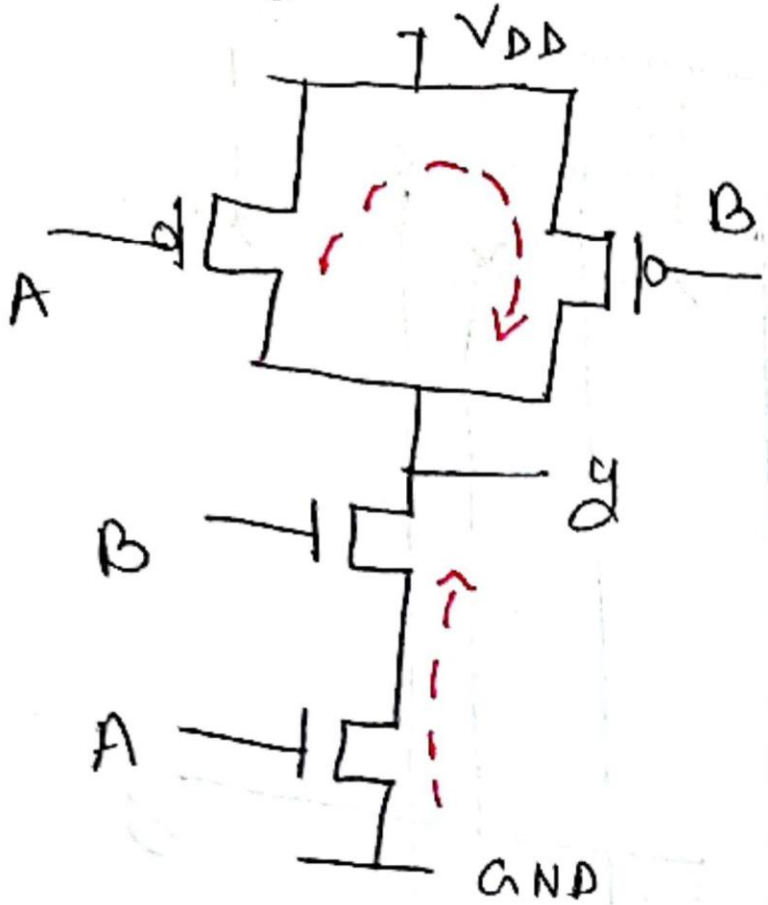
Stick Diagram (CMOS): Basic Steps

- Steps for CMOS are similar to NMOS
- But one difference is that depletion mode FETs are not used.
- Here, yellow/ brown is used to identify PMOS.
- The two types of FET, n and p, are separated in the diagram by the demarcation line.
- This line represents the well (n/p-well).
- Above this line are all p-type MOSFETs.
- Below this line n-type MOSFETs are present.

CMOS Invertor (Stick Diagram)

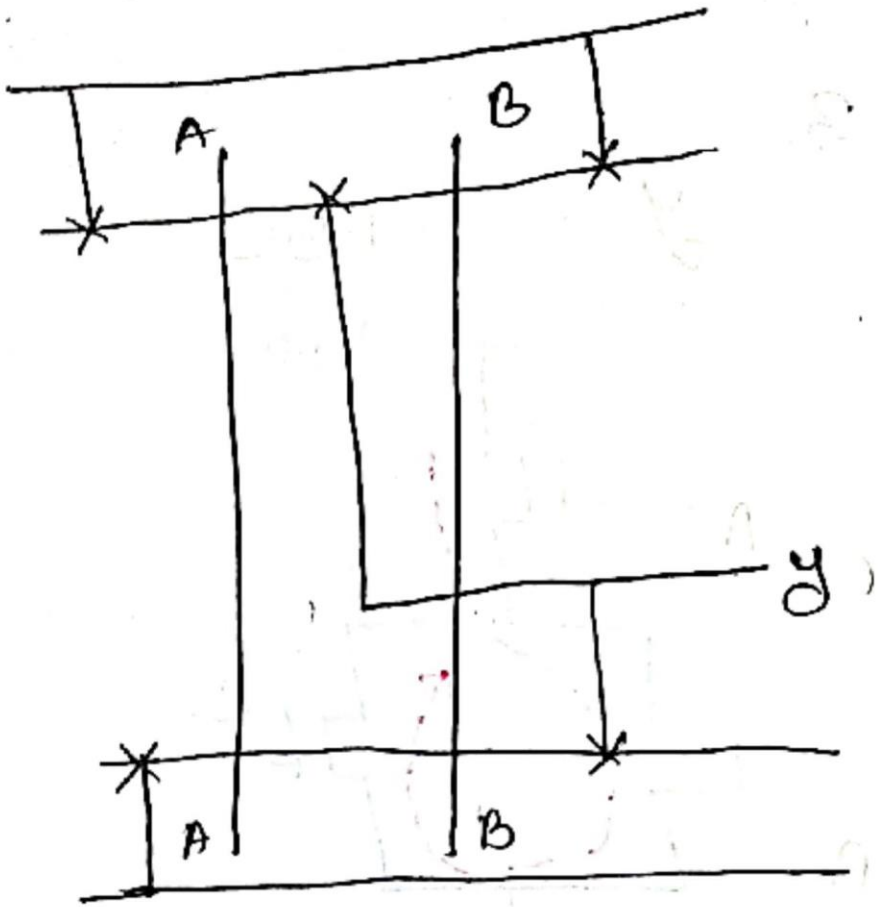


$y = \overline{AB}$

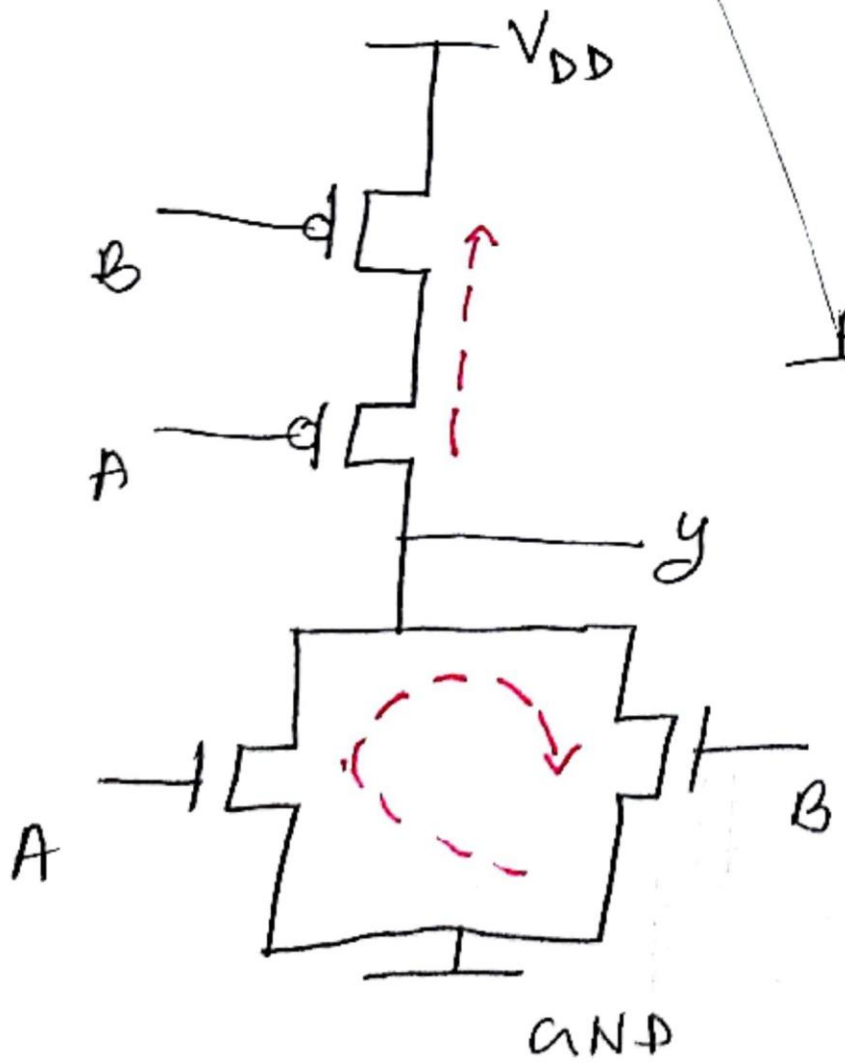


Path

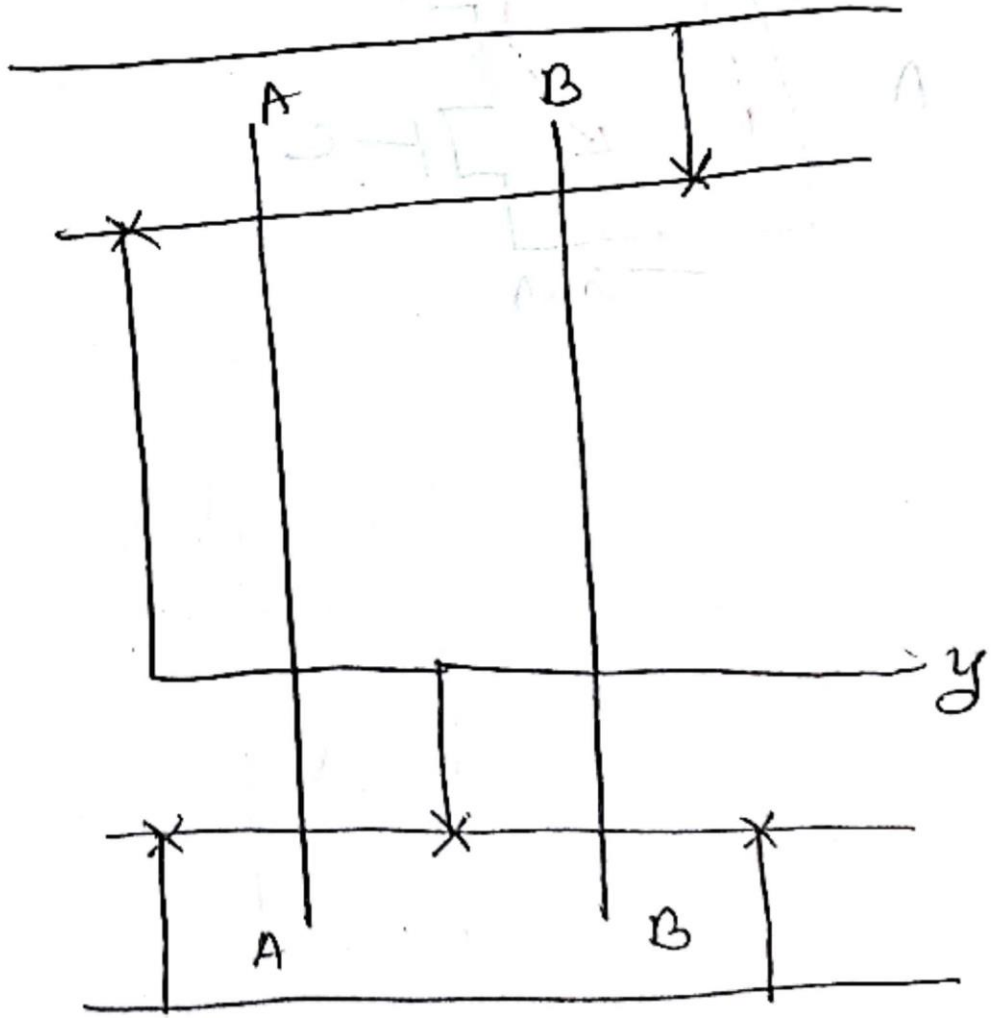
AB



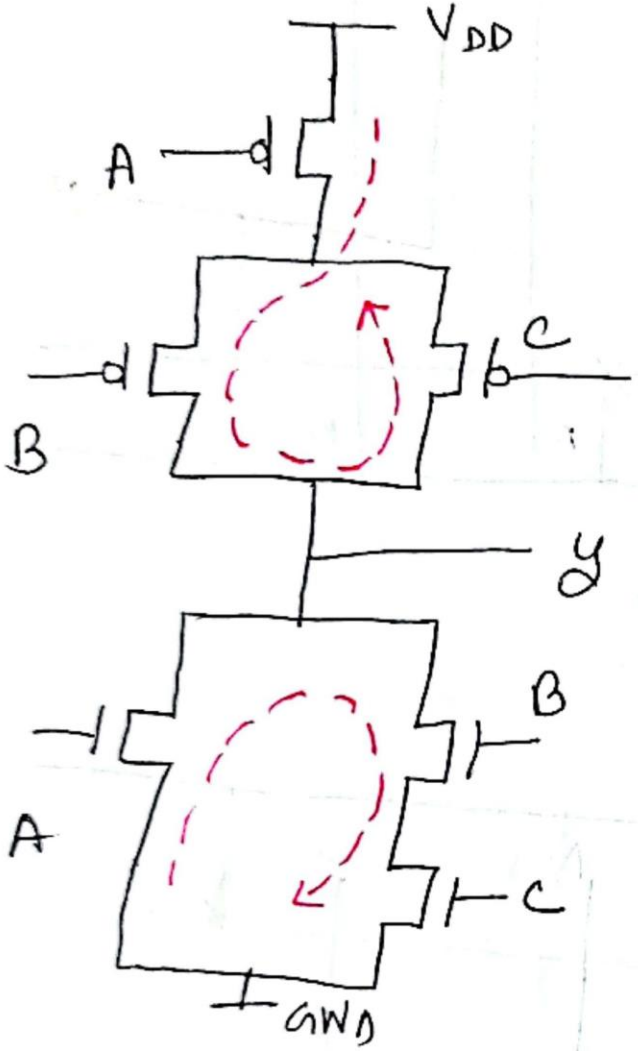
$$y = \overline{A+B}$$



path
~~AB~~

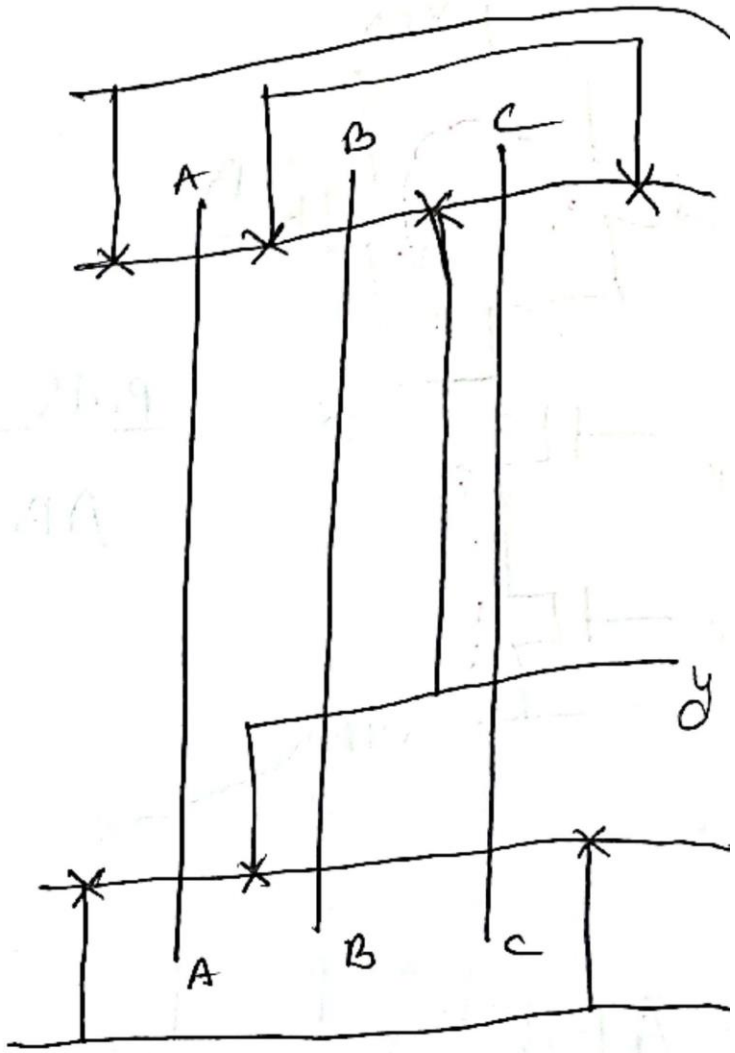


3. $Y = \overline{A + BC}$



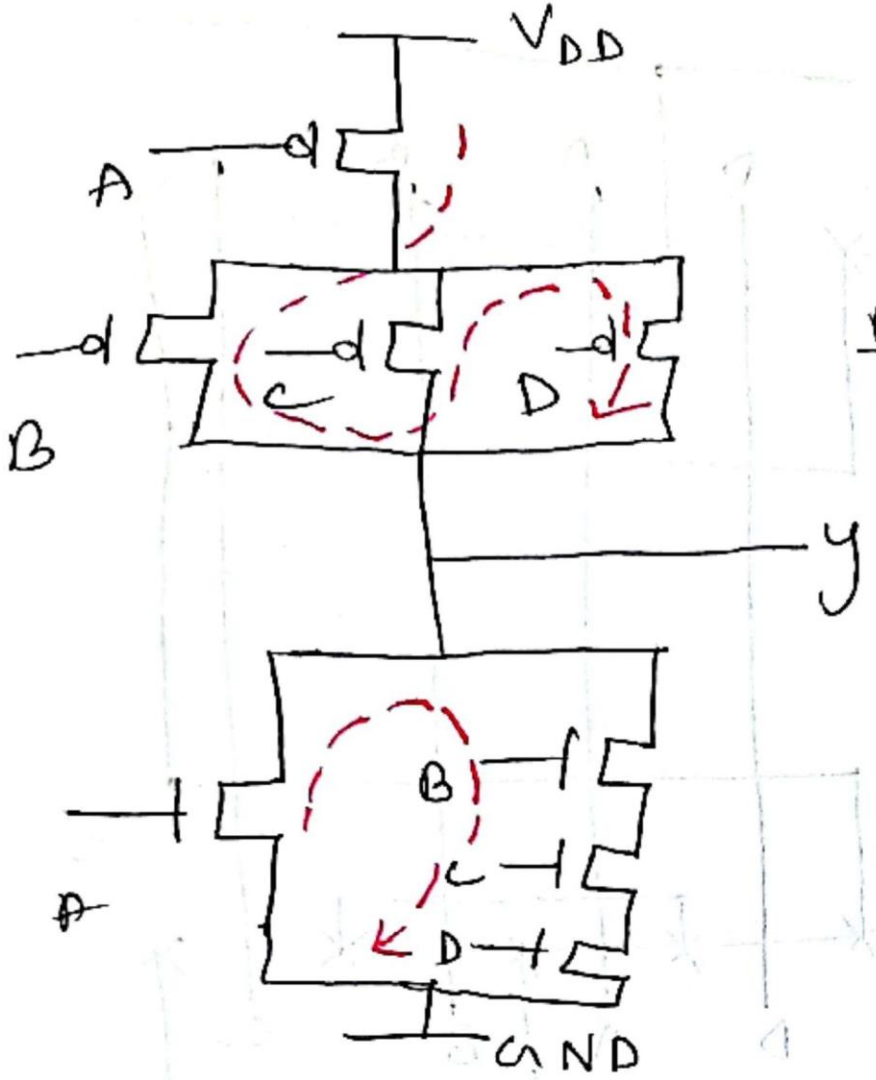
Path

ABc

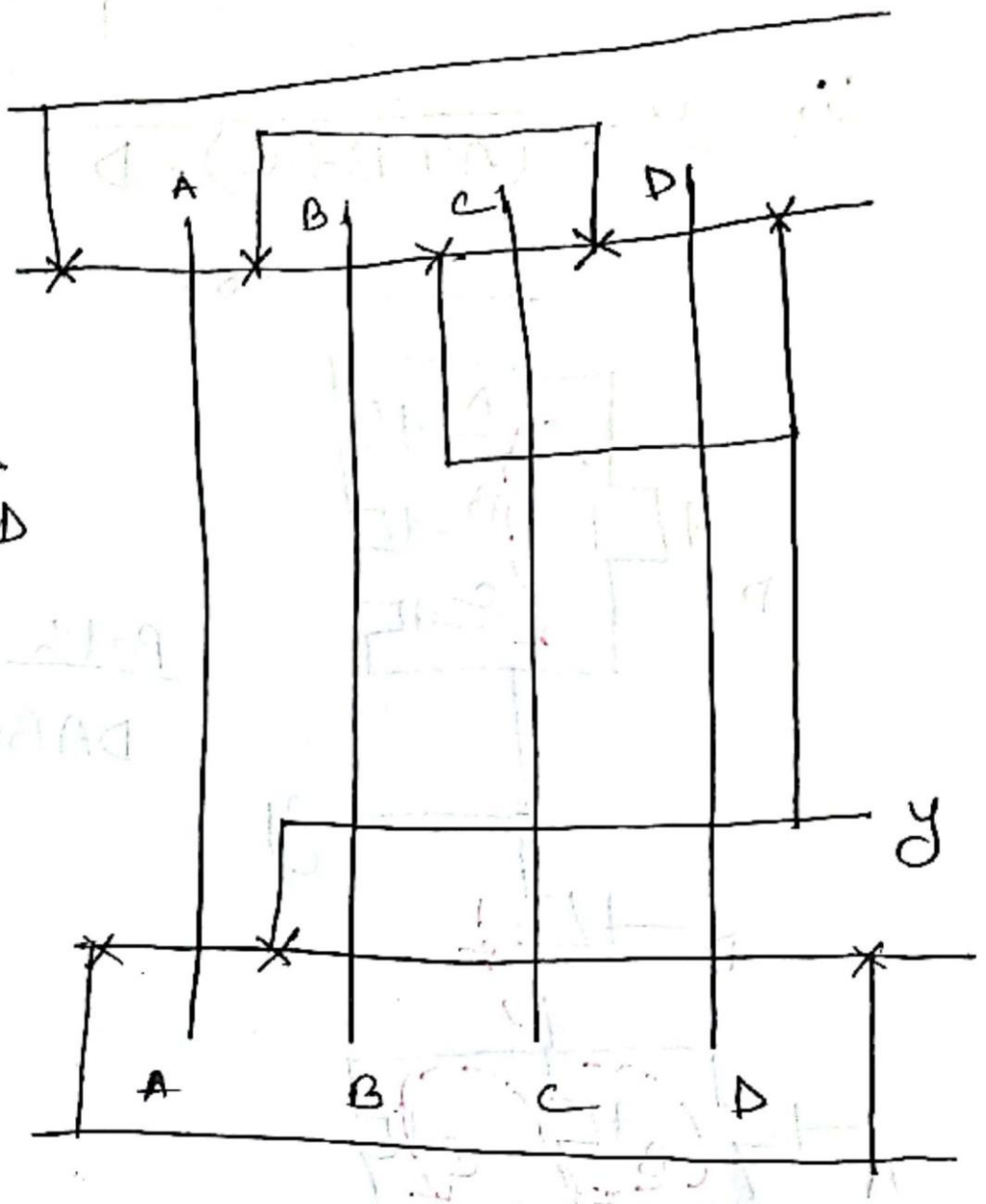


4.

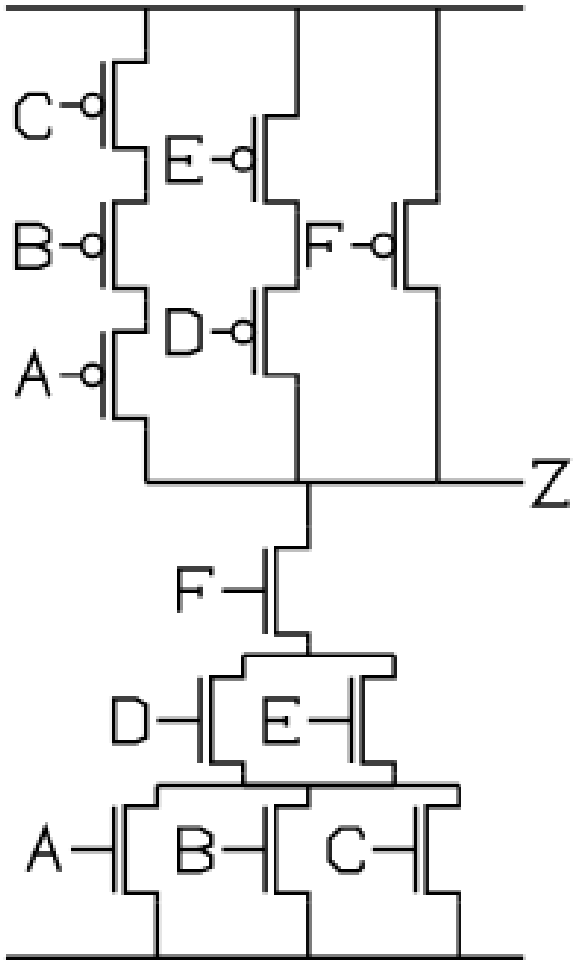
$$y = \overline{A + BCD}$$



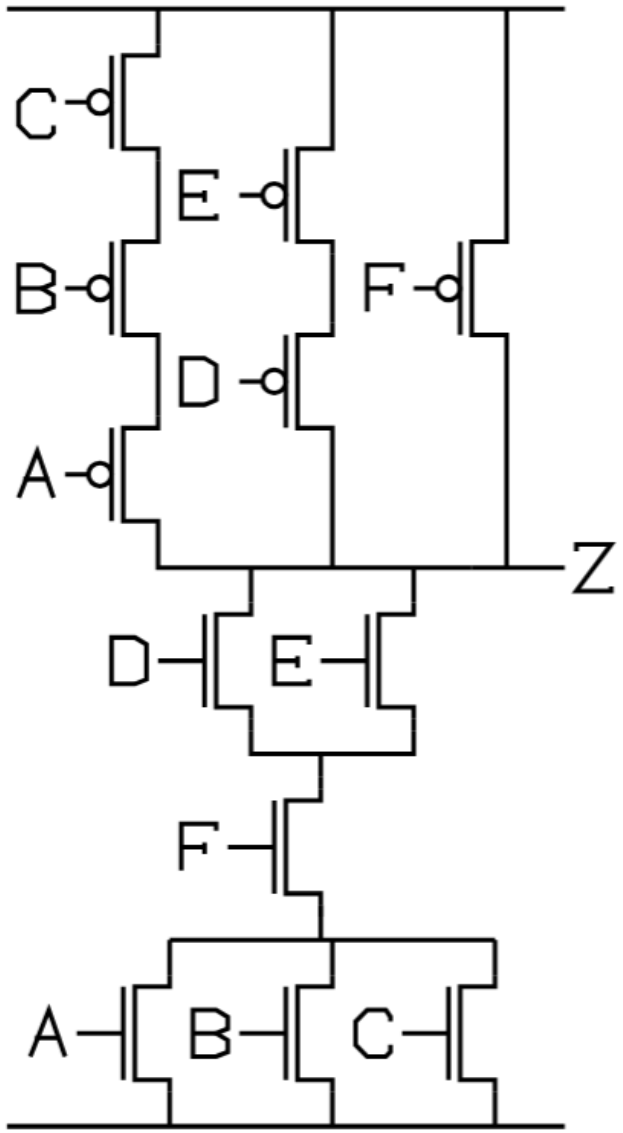
path
ABCD



$$Z = \overline{(A+B+C) \bullet (D+E) \bullet F}$$

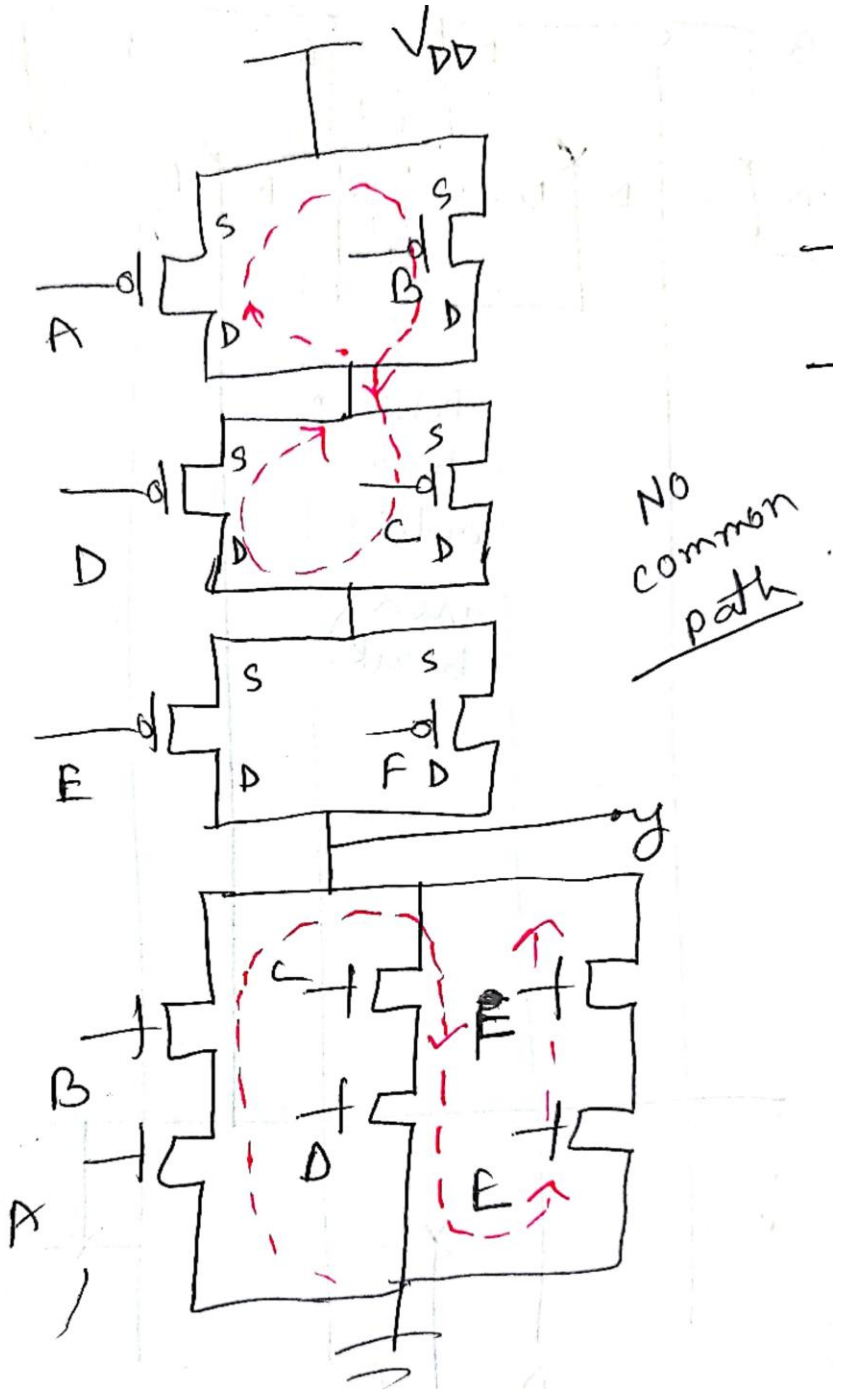


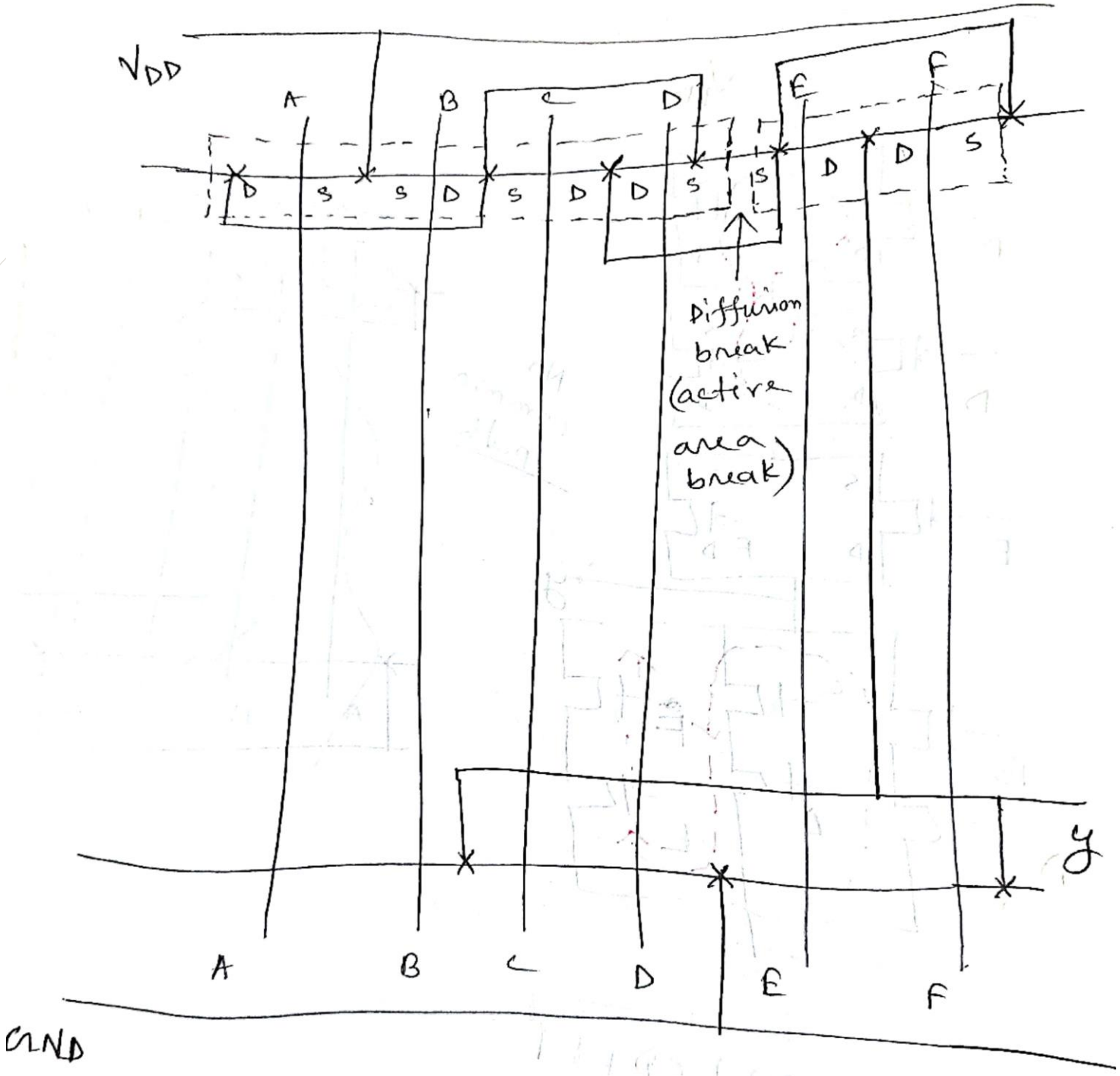
Does Not maintain Euler path



$$Z = \overline{(A+B+C) \bullet (D+E) \bullet F}$$

$y=\overline{AB+BD+EF)}$




VDD
A
B
C
D
E
F
D
S
S
D
S
D
D
S
S
D
D
S
Diffusion break (active area break)
y
A
B
C
D
E
F
GND